

Analytical Fragmentation Predictions for vAttention Cache Layouts

1 Goal

This note records the analytical fragmentation formulas that match the metric plotted in our empirical figures. The important distinction is that the runtime fragmentation percentage is a *tail-slack percentage in token capacity*, not the older proposal’s fixed-byte-waste ratio.

2 Runtime Fragmentation Metric

Let

- C be the current context length in tokens,
- T be the allocator’s number of cached tokens per physical page,
- $B(C) = \lceil C/T \rceil$ be the number of mapped blocks needed for that context.

Then the exact fragmentation percentage used by the runtime is

$$F_{\text{exact}}(C) = 100 \cdot \frac{B(C)T - C}{B(C)T} = 100 \cdot \frac{\lceil C/T \rceil T - C}{\lceil C/T \rceil T}. \quad (1)$$

This produces the sawtooth pattern observed in the empirical plots.

3 Worst-Case Tail-Waste Envelope

For a smooth upper envelope, consider the point immediately after a new page has been allocated. At the beginning of each tooth, one page has just been added and is almost entirely empty. This gives the smooth worst-case envelope

$$F_{\text{worst}}(C) \approx 100 \cdot \frac{T}{C + T}. \quad (2)$$

This is the dotted curve we overlay on the empirical graphs. Its purpose is not to reproduce every point inside the tooth, but to predict the vertical scale of the fragmentation peaks.

4 Average Tail-Waste Approximation

If we assume that the final mapped page is half full on average, then the average smooth approximation is

$$F_{\text{avg}}(C) \approx 100 \cdot \frac{T/2}{C + T/2}. \quad (3)$$

This is closer in spirit to the original proposal, but it does not emphasize the peak heights as clearly as the worst-case envelope.

5 Architecture-Dependent Tokens Per Page

The architecture-specific part of the prediction is the number of cached tokens per page:

$$T = \left\lfloor \frac{P}{S_{\text{page-token}}} \right\rfloor, \quad (4)$$

where P is the physical page size in bytes and $S_{\text{page-token}}$ is the per-token page-buffer footprint used by the runtime.

5.1 Dense KV Layouts: MHA, GQA, and MQA

For dense KV cache layouts, the runtime stores one key slice and one value slice per layer, but the page-buffer token size is computed per layer per side:

$$S_{\text{page-token}}^{\text{dense}} = H_{\text{kv,loc}} \cdot D \cdot b, \quad (5)$$

where

- $H_{\text{kv,loc}}$ is the number of local KV heads on one tensor-parallel worker,
- D is the head dimension,
- b is the number of bytes per element.

Therefore

$$T_{\text{dense}} = \left\lfloor \frac{P}{H_{\text{kv,loc}} D b} \right\rfloor. \quad (6)$$

This gives

$$F_{\text{worst}}^{\text{dense}}(C) \approx 100 \cdot \frac{\left\lfloor \frac{P}{H_{\text{kv,loc}} D b} \right\rfloor}{C + \left\lfloor \frac{P}{H_{\text{kv,loc}} D b} \right\rfloor}. \quad (7)$$

MHA. For MHA, all attention heads are KV heads, so $H_{\text{kv,loc}} = H_{\text{loc}}$. The proposal was directionally correct that MHA uses the full head count, but the plotted fragmentation percentage is controlled by T_{dense} , not by the total cache bytes per token across all layers.

GQA. For GQA, $H_{\text{kv,loc}}$ is smaller than in MHA, which increases T and lowers the fragmentation envelope.

5.2 MLA Layouts

For MLA, the runtime stores the latent KV state and the RoPE key component. The per-token page-buffer footprint is

$$S_{\text{page-token}}^{\text{MLA}} = (r_{\text{kv}} + d_{\text{rope}}) \cdot b, \quad (8)$$

where

- r_{kv} is the KV latent rank,
- d_{rope} is the RoPE key dimension.

Therefore

$$T_{\text{MLA}} = \left\lfloor \frac{P}{(r_{\text{kv}} + d_{\text{rope}})b} \right\rfloor, \quad (9)$$

and

$$F_{\text{worst}}^{\text{MLA}}(C) \approx 100 \cdot \frac{\left\lfloor \frac{P}{(r_{\text{kv}} + d_{\text{rope}})b} \right\rfloor}{C + \left\lfloor \frac{P}{(r_{\text{kv}} + d_{\text{rope}})b} \right\rfloor}. \quad (10)$$

6 Concrete Predictions for Current Runs

All current runs use $P = 2,097,152$ bytes (2 MiB pages).

Run	Architecture	T	Worst-case envelope
Qwen-14B	MHA / dense KV	819	$F_{\text{worst}}(C) \approx 100 \cdot \frac{819}{C+819}$
Mistral-Nemo-12B	GQA / dense KV	4096	$F_{\text{worst}}(C) \approx 100 \cdot \frac{4096}{C+4096}$
DeepSeek-V2-Lite	MLA	1820	$F_{\text{worst}}(C) \approx 100 \cdot \frac{1820}{C+1820}$
Mistral-Nemo-12B (Synthetic MLA)	MLA	5461	$F_{\text{worst}}(C) \approx 100 \cdot \frac{5461}{C+5461}$

Table 1: Worst-case fragmentation envelopes for the current experimental runs.

7 Relation to the Original Proposal

The original proposal modeled a fixed average byte waste that scales with the number of fragmentation points. That model is still useful for reasoning about aggregate wasted bytes. However, the plotted metric in the current experiments is different:

- it is measured as tail slack in mapped token capacity,
- it depends directly on T , the tokens per page,
- and it naturally produces a sawtooth curve rather than a single smooth amortization curve.

For the current figures, the right analytical objects are therefore:

- the exact sawtooth law $F_{\text{exact}}(C)$, and
- the worst-case smooth envelope $F_{\text{worst}}(C)$.